

HR ANALYTICS PROJECT REPORT

Author: Bhosale Saurav

Database: HRAalyticsDB

Tool Used: MySQL

Date: July 2026

1. Project Overview

This project is a **complete HR analytics solution built entirely using SQL**.

It analyzes employee data to understand **attrition patterns, salary distribution, job satisfaction, and workforce structure**.

The project demonstrates the use of:

- SQL Aggregations
 - Window Functions
 - Subqueries
 - Views
 - Stored Procedures
 - KPI-based reporting logic
-

2. Database Setup

- Database: HRAalyticsDB
- Main Table: EmployeeData (renamed from raw dataset)

The dataset includes employee attributes such as:

- Department, Job Role
 - Monthly Income
 - Attrition status
 - Work-life balance
 - Job satisfaction
 - Performance rating
 - Experience details
-

3. SQL Analysis Performed

A. Employee Overview

- Total employees count
- Gender distribution
- Age statistics (min, max, average)
- Department and job role distribution

📌 Insight: Helps understand workforce composition.

◆ B. Salary Analysis

- Average salary by department
- Average salary by job role
- Top 10 highest-paid employees
- Employees earning above company average

📌 Insight: Salary varies significantly across roles and departments.

◆ C. Attrition Analysis (Core Focus)

- Overall attrition rate
- Attrition by department
- Attrition by job role
- Attrition vs overtime
- Attrition vs salary
- Attrition vs satisfaction

📌 Key Finding:
Overtime, low salary, and poor satisfaction strongly increase attrition risk.

◆ D. Work Experience Analysis

- Years at company by department
- Employees with long tenure
- Employees never promoted
- Experience comparison across departments

📌 Insight:
Some departments retain employees longer than others.

◆ E. Employee Well-being Analysis

- Job satisfaction by department
- Work-life balance by department
- Employees with high risk (low satisfaction + overtime)

📌 Insight:

Poor work conditions directly impact retention.

🧠 4. Advanced SQL Analysis

◆ Window Functions Used:

- Ranking employees by salary
- Top 3 salaries per department
- Running total of salary by department
- Department-wise salary comparison
- Highest-paid employee per department

📌 Insight:

Helps analyze internal hierarchy and fairness.

🚨 5. Attrition Risk Detection Logic

Employees identified as high risk:

- Working overtime
- Low work-life balance (≤ 2)
- Low job satisfaction (≤ 2)

📌 Insight:

Combination of workload + dissatisfaction strongly predicts attrition.

📊 6. HR KPIs Calculated

Company-Level KPIs:

- Total employees
- Retention rate
- Attrition rate
- Average salary

- Average tenure

Department-Level KPIs:

- Employee distribution
- Attrition rate by department
- Average salary per department
- Average experience

Job Role KPIs:

- Attrition rate by job role
 - Salary comparison across roles
-

7. SQL Views Created (Reusable Data Layer)

To support reporting, the following views were created:

- employee_summary
- department_dashboard
- jobrole_dashboard
- salary_details1

 Purpose:

These views act as a **mini data warehouse layer** for reporting.

8. Stored Procedures Created

- Department-wise HR summary procedure
- Salary filter procedure (employees above threshold)

 Purpose:

Enables dynamic reporting using input parameters.

9. Key Business Insights

Major Attrition Drivers:

- OverTime (strongest factor)
- Low monthly income
- Low job satisfaction
- Poor work-life balance

- Short tenure

Secondary Factors:

- Department differences
 - Job role differences
 - Experience level
-

10. Final Conclusion

This project demonstrates a **complete HR analytics system built purely using SQL**, without any external BI tools.

It successfully:

- ✓ Analyzes employee structure
 - ✓ Identifies attrition patterns
 - ✓ Evaluates salary fairness
 - ✓ Measures employee satisfaction
 - ✓ Builds reusable reporting views
 - ✓ Implements procedural reporting logic
-

11. Resume Summary (SQL Project)

Built a full-scale HR Analytics system using SQL, analyzing employee attrition, salary trends, job satisfaction, and workforce distribution. Developed complex queries using window functions, stored procedures, and views to generate actionable HR insights and KPIs.
